

TOYAMA, Japan

WMO: 476070

Lat: 36.7094N

Lon: 137.2030E

Elev: 57

StdP: 14.67

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.7	29.3	20.1	15.1	37.1	22.3	16.8	36.8	20.9	45.4	19.0	44.2	6.4	200	0.417

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.2	93.1	78.0	90.9	77.5	88.4	76.7	80.1	88.6	79.0	87.0	78.0	85.5	9.8	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.9	145.4	84.6	76.8	140.1	83.6	75.8	135.1	82.5	43.8	89.2	42.6	87.2	41.5	85.8	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.2	17.1	14.5	DB	24.4	97.8	2.1	2.2	22.9	99.4	21.6	100.7	20.4	101.9	18.9	103.5
			WB	23.1	81.7	2.0	1.5	21.7	82.8	20.5	83.7	19.4	84.5	18.0	85.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.6	37.5	38.4	44.7	54.4	64.2	70.9	78.9	81.3	73.6	63.2	52.4	42.6	
	DBStd	15.96	4.49	5.27	6.27	6.62	5.44	4.40	4.84	4.33	5.28	5.27	5.94	5.64	
	HDD50	1214	389	328	189	29	0	0	0	0	0	0	40	239	
	HDD65	3805	853	745	628	323	80	5	0	0	2	96	378	695	
	CDD50	4364	0	3	26	161	441	627	895	972	707	410	113	9	
	CDD65	1479	0	0	0	5	56	181	430	507	259	40	1	0	
	CDH74	12184	0	0	1	51	349	856	3822	5312	1647	144	2	0	
	CDH80	4293	0	0	0	8	58	179	1422	2145	459	22	0	0	
Wind	WSAvg	6.4	6.3	6.3	7.3	7.4	6.9	5.7	5.9	6.0	6.0	6.2	6.5	6.6	
Precipitation	PrecAvg	87.5	9.6	6.3	5.9	5.0	5.0	6.6	8.7	7.3	8.1	6.3	8.5	10.3	
	PrecMax	105.2	16.5	11.1	8.7	9.3	12.2	12.4	20.6	19.8	15.2	12.0	16.1	21.8	
	PrecMin	60.1	4.7	2.4	2.7	1.5	2.0	2.3	1.3	1.6	3.2	3.4	5.6	6.0	
	PrecStd	11.0	2.7	2.1	1.7	2.1	2.6	2.5	5.1	4.4	3.1	1.9	2.2	3.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.7	61.9	70.4	79.8	85.5	88.6	95.5	96.4	92.4	83.0	72.8	63.7	
		MCWB	45.6	52.5	54.1	59.3	65.3	73.9	77.5	77.7	76.7	69.0	59.7	53.6	
	2%	DB	50.4	55.0	64.5	73.8	81.2	84.4	92.6	93.8	88.3	77.7	68.7	58.0	
		MCWB	43.8	46.7	51.1	58.2	63.1	70.8	77.5	78.2	75.4	66.5	58.7	50.6	
	5%	DB	47.5	50.8	60.0	70.0	77.9	81.7	90.2	91.8	85.1	74.5	65.4	54.4	
		MCWB	42.0	44.0	49.5	56.3	62.3	70.1	77.2	78.2	74.3	65.2	56.7	48.5	
	10%	DB	44.9	47.3	55.8	66.1	74.7	78.9	87.5	89.5	82.1	71.8	62.2	51.5	
		MCWB	40.5	42.0	47.5	54.7	61.4	69.1	76.5	77.8	73.2	63.8	55.7	46.6	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.2	53.7	56.7	64.1	69.6	76.4	80.9	82.2	79.4	71.7	63.1	56.2
			MCDB	52.7	61.1	65.2	74.2	77.0	83.9	90.1	90.2	86.8	77.7	67.7	60.5
2%		WB	44.9	48.0	53.2	60.5	66.9	74.5	79.5	80.8	77.4	69.2	60.6	52.0	
		MCDB	49.0	53.5	61.3	69.6	75.4	80.4	88.2	89.3	84.1	74.7	66.2	56.1	
5%		WB	42.8	45.0	50.6	58.3	65.1	72.7	78.5	79.7	75.9	67.3	58.4	49.3	
		MCDB	46.5	49.8	57.5	66.7	73.0	78.4	86.7	87.8	82.2	72.1	63.4	53.4	
10%		WB	40.9	42.2	48.4	56.2	63.5	71.1	77.4	78.7	74.5	65.4	56.3	47.0	
		MCDB	44.4	46.7	54.7	63.6	71.3	76.5	85.1	86.4	80.3	70.1	61.2	50.9	
Mean Daily Temperature Range		5% DB	MDBR	9.8	11.7	15.2	16.8	16.0	12.8	12.3	13.2	12.7	13.9	13.5	11.1
			MCDBR	14.7	18.6	22.8	23.7	21.8	17.4	16.5	16.3	16.3	16.7	18.1	16.5
	MCWBR		9.9	12.4	13.3	11.8	9.5	7.5	6.3	6.7	7.0	8.4	10.7	10.8	
	5% WB	MCDBR	13.3	17.2	19.3	19.0	17.4	14.0	14.8	14.6	14.1	13.4	15.1	14.9	
		MCWBR	9.6	12.4	12.5	10.8	9.1	6.8	6.3	6.6	6.8	7.6	9.9	10.9	
Clear-Sky Solar Irradiance	taub	0.411	0.440	0.504	0.531	0.534	0.557	0.545	0.531	0.478	0.441	0.418	0.410		
	taud	1.963	1.905	1.835	1.870	1.929	1.951	2.048	2.091	2.199	2.197	2.158	2.038		
	Ebn at Noon	232	243	240	242	244	238	240	240	246	243	232	223		
	Edh at Noon	44	52	62	63	60	59	53	50	43	39	36	38		
All-Sky Solar Radiation	RadAvg	523	785	1119	1448	1665	1572	1501	1551	1220	983	656	455		
	RadStd	54	63	88	119	180	116	227	166	93	85	56	59		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (23 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page